FACULDADE DE INFORMÁTICA E ADMINISTRAÇÃO PAULISTA: ENGENHARIA DE DADOS

Graph Databases and Analytics

SÃO PAULO
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FIAP

29ABD

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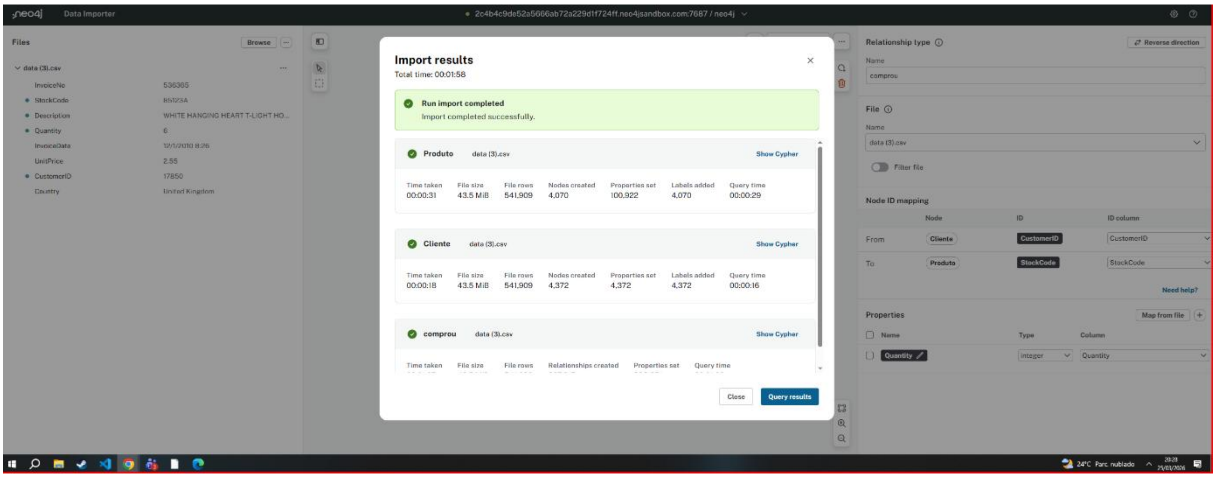
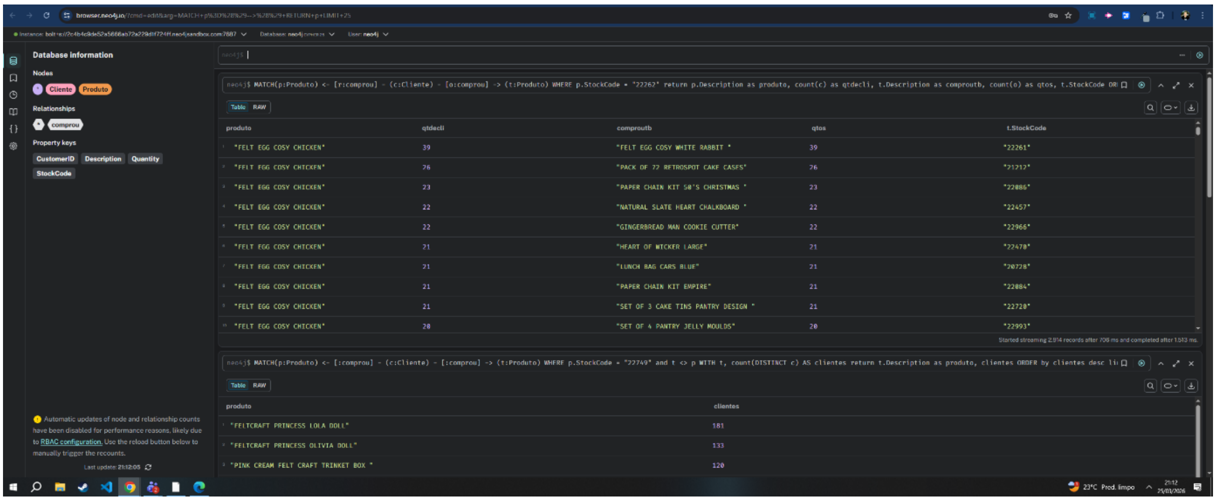
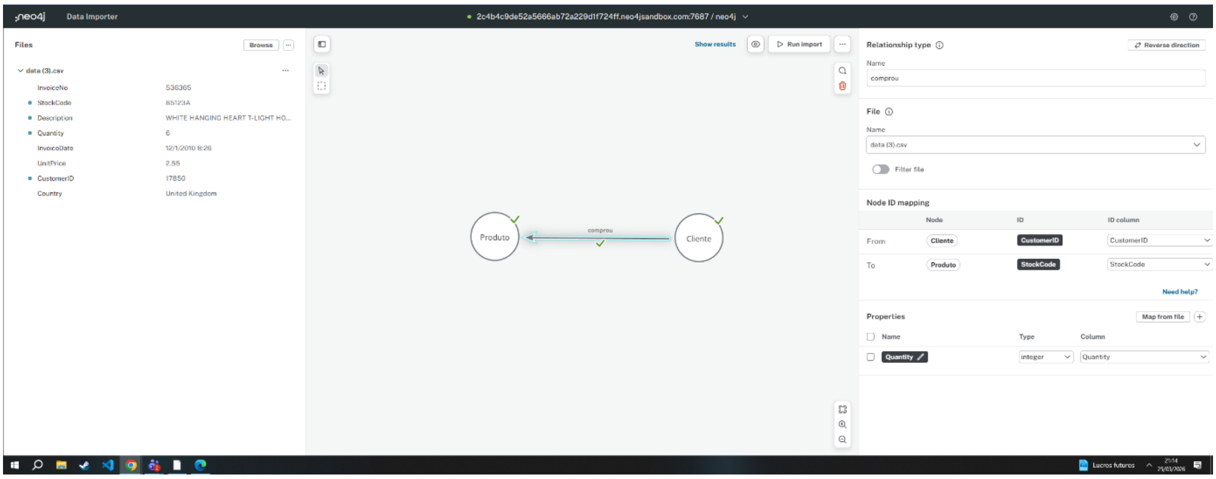
Exercício1 –Construindo grafos

Construir algumas consultas para recomendação de produtos a partir







a) do produto mais vendido

```
neo4j$ MATCH (c:Cliente)-[:comprou]->(p:Produto) WITH p, count(*) AS vendas RETURN p.StockCode AS codigo_produto, p.Description AS produto, vendas ORDER BY vendas DESC LIMIT 1
```

codigo_produto	produto	vendas
"22423"	"REGENCY CAKESTAND 3 TIER"	887

Started streaming 1 record after 255 ms and completed after 948 ms.

b) de um cliente qualquer

```
MATCH (c1:Cliente {CustomerID:17850})-[:comprou]->(p:Produto)
MATCH (c2:Cliente)-[:comprou]->(p)
WHERE c1 <> c2
MATCH (c2)-[:comprou]->(rec:Produto)
WHERE NOT (c1)-[:comprou]->(rec)
WITH rec, count(DISTINCT c2) AS recomendacoes
RETURN rec.StockCode AS codigo_produto,
       rec.Description AS produto_recomendado,
       recomendacoes
ORDER BY recomendacoes DESC
LIMIT 10
```

codigo_produto	produto_recomendado	recomendacoes
"22423"	"REGENCY CAKESTAND 3 TIER"	579
"47566"	"PARTY BUNTING"	496
"84879"	"ASSORTED COLOUR BIRD ORNAMENT"	480
"85099B"	"JUMBO BAG RED RETROSPOT"	479
"22720"	"SET OF 3 CAKE TINS PANTRY DESIGN "	439
"22469"	"HEART OF WICKER SMALL"	426
"22457"	"NATURAL SLATE HEART CHALKBOARD "	417
"22086"	"PAPER CHAIN KIT 50'S CHRISTMAS "	415
"23298"	"SPOTTY BUNTING"	411
"21212"	"PACK OF 72 RETROSPOT CAKE CASES"	408

Started streaming 10 records after 52 ms and completed after 5.364 ms.

c) de outro produto qualquer

```
neo4j$ MATCH (p:Produto {StockCode:"85123A"})<-[:comprou]-(c:Cliente)-[:comprou]->(t:Produto) WHERE t <> p WITH t, count(DISTINCT c) AS clientes RETURN t.StockCode AS codigo_produto, t.Description AS produto_recomendado, cli
```

codigo_produto	produto_recomendado	clientes
"21733"	"RED HANGING HEART T-LIGHT HOLDER"	312
"22469"	"HEART OF WICKER SMALL"	266
"22470"	"HEART OF WICKER LARGE"	266
"22423"	"REGENCY CAKESTAND 3 TIER"	262
"47566"	"PARTY BUNTING"	255
"22457"	"NATURAL SLATE HEART CHALKBOARD "	250
"84879"	"ASSORTED COLOUR BIRD ORNAMENT"	238
"22804"	"PINK HANGING HEART T-LIGHT HOLDER"	230
"85099B"	"JUMBO BAG RED RETROSPOT"	222
"82482"	"WOODEN PICTURE FRAME WHITE FINISH"	216

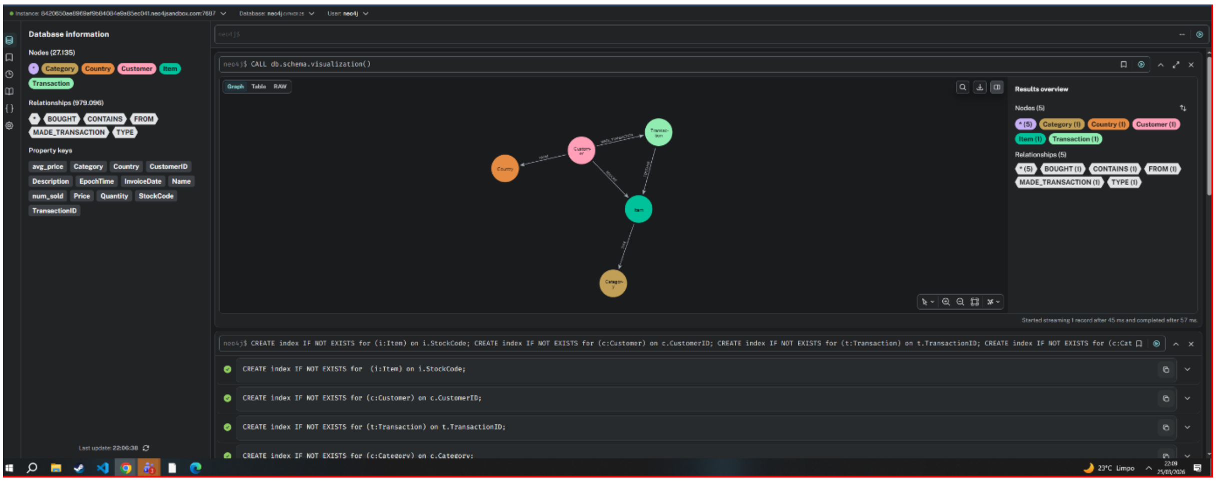
Started streaming 10 records after 597 ms and completed after 1.007 ms.

Exercício 2: Case Varejo - Hands on aula 2 - página 14 - responder questões grifadas

Criar uma base de dados a partir do script load_data_transacoes.txt

Após carga verificar o modelo

CALL db.schema.visualization()



1) Explorar os dados respondendo:

Total de itens, de categorias, de clientes, de países e de transações

Quais os dez clientes com o maior número de transações e quais os produtos mais comprados?

Database: neo4j CYPHER 25 User: neo4j

```
MATCH (c:Customer)-[:MADE_TRANSACTION]->(t:Transaction)
WITH c, count(t) AS total_transacoes
RETURN c.CustomerID AS cliente,
       total_transacoes
ORDER BY total_transacoes DESC
LIMIT 10
```

cliente	total_transacoes
12748	236
14911	225
17841	145
13089	132
15311	121
14606	120
12971	113
14646	102
16029	91
13408	89

Started streaming 10 records after 103 ms and completed after 186 ms.

```
neo4j$ MATCH (c:Customer)-[b:BOUGHT]->(i:Item) WITH i, sum(b.Quantity) AS total_vendas RETURN i.StockCode AS codigo_produto, i.Description AS produto, total_vendas ORDER BY total_vendas DESC LIMIT 10
```

codigo_produto	produto	total_vendas
23843	"PAPER CRAFT LITTLE BIRDIE"	80995
23166	"MEDIUM CERAMIC TOP STORAGE JAR"	78033
22197	"POPCORN HOLDER"	56921
84077	"WORLD WAR 2 GLIDERS ASSTD DESIGNS"	55047
84879	"ASSORTED COLOUR BIRD ORNAMENT"	36461
21212	"PACK OF 72 RETROSPOT CAKE CASES"	36419
23084	"RABBIT NIGHT LIGHT"	30788
22492	"MINI PAINT SET VINTAGE "	26633
22616	"PACK OF 12 LONDON TISSUES "	26135
21977	"PACK OF 60 PINK PAISLEY CAKE CASES"	24854

Started streaming 10 records after 84 ms and completed after 3.021 ms.

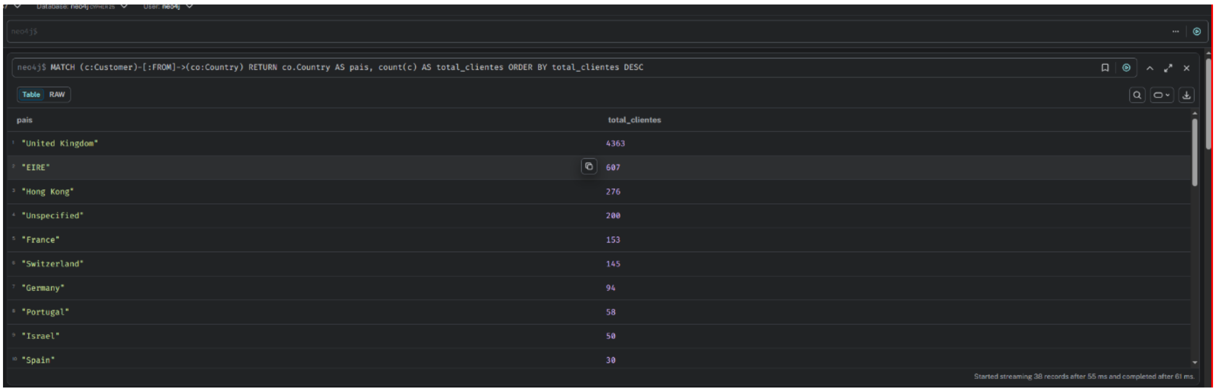
Total de itens por categoria

```
neo4j$ MATCH (i:Item)-[:TYPE]->(c:Category) RETURN c.Category AS categoria, count(i) AS total_itens ORDER BY total_itens DESC
```

categoria	total_itens
"HOME DÉCOR"	866
"KITCHEN"	593
"STATIONERY"	312
"TOY"	244
"ACCESSORIES"	214
"SEASONAL"	202
"GARDEN"	80
"PARTY"	79
"JEWELRY"	68
"ARTS AND CRAFTS"	66

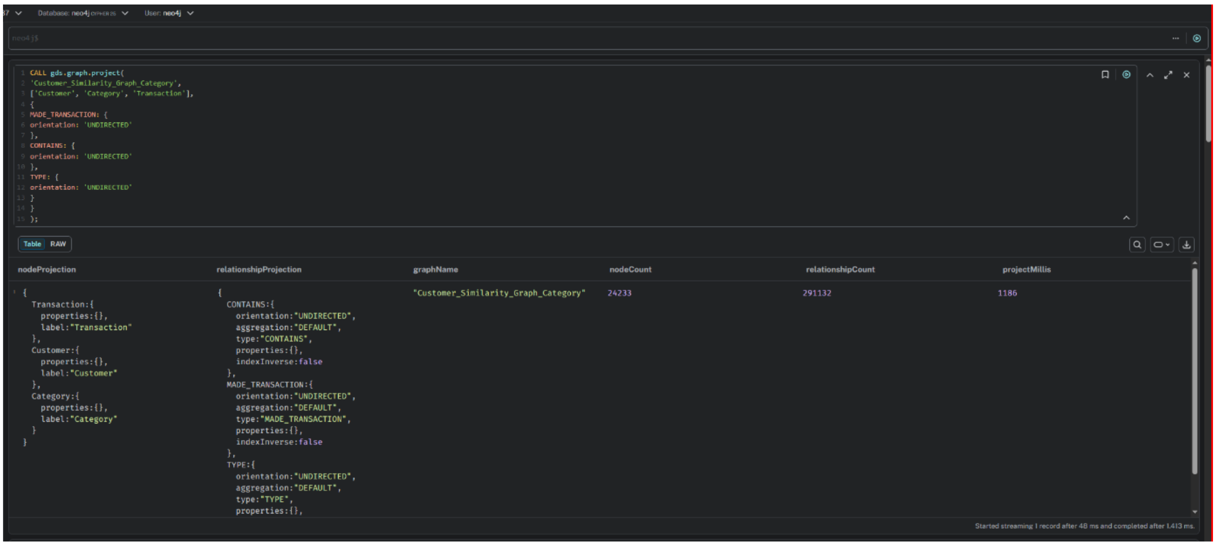
Started streaming 12 records after 81 ms and completed after 92 ms.

Total de clientes por país

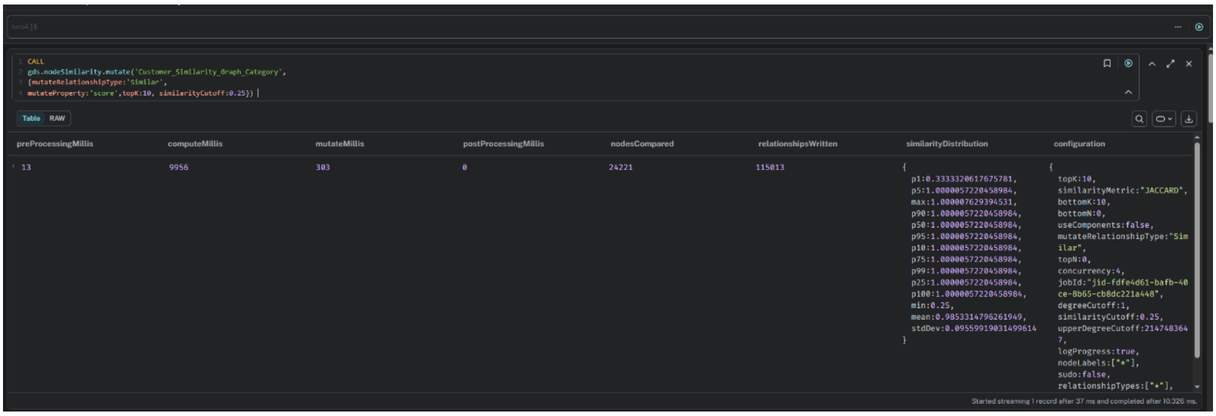


2. Realizar Segmentação de Clientes

a. Similaridade por Cliente por Categoria – criar projeção Customer_Similarity_Graph_Category



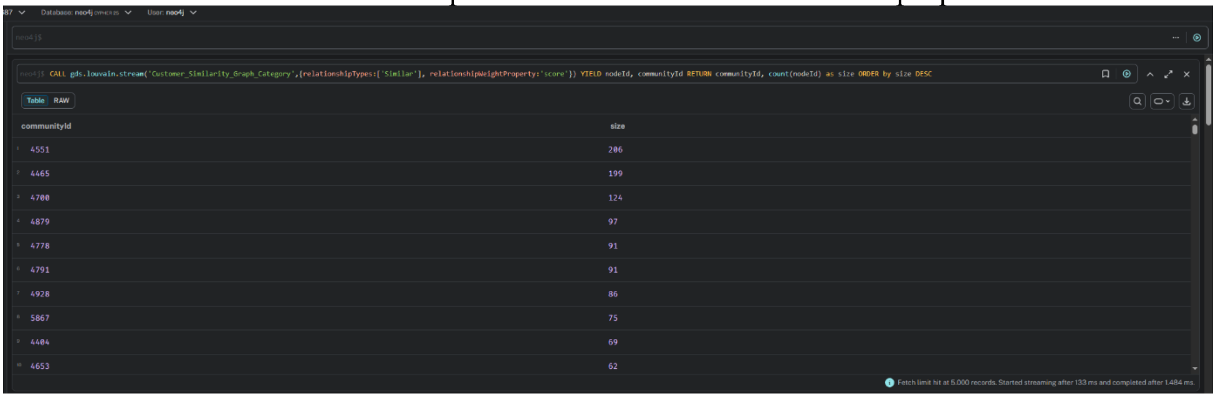
b. Similaridade por Cliente por Categoria – gravando relacionamento Similar



2) Explicar o comando mutate e destacar os dez clientes relacionados

O comando Mutate vai gravar o resultado temporariamente na projeção de grafo, mas sem persistir os dados diretamente no banco.

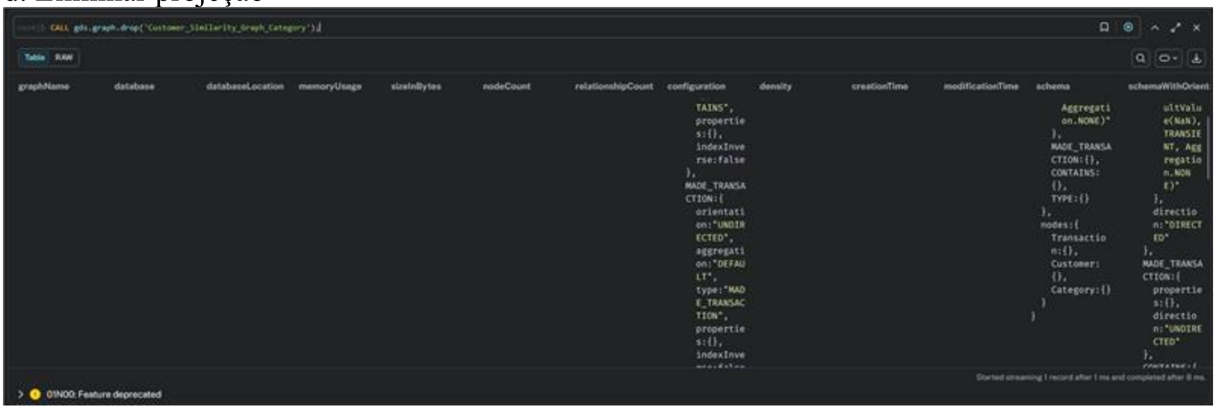
c. Louvain comunidade formada a partir do relacionamento Similar e propriedade score



communityId	size
4551	206
4465	199
4700	124
4879	97
4778	91
4791	91
4928	86
5867	75
4404	69
4653	62

3) Quantas comunidades foram criadas com o total de clientes em cada uma?

d. Eliminar projeção



e. Cliente similaridade por item

f. Node similaridade

```
CALL gds.nodeSimilarity.mutate(
  'Customer_Similarity_Graph_Items',
  {
    mutateRelationshipType: 'Similar',
    mutateProperty: 'score',
    topK: 10,
    similarityCutoff: 0.1
  }
);
```

Table RAW

preProcessingMillis	computeMillis	mutateMillis	postProcessingMillis	nodesCompared	relationshipsWritten	similarityDistribution	configuration
0	3168	16	0	4371	17298	{ p1:0.09999990463256836, p5:0.10119009017944336, max:0.5080223083496093, p90:0.16959047317504883, p50:0.12209272384643555, p95:0.18762445449829102, p10:0.10285711288452148, p75:0.14328336715698242, p99:0.2308802604675293, p25:0.1088705062866211, p100:0.5080218315124512, min:0.09999990463256836,	{ topK:10, similarityMetric:"JACCARD", bottomK:10, bottomN:0, useComponents:false, mutateRelationshipType:"Similar", topN:0, concurrency:4, jobId:"jid-24c3fb79-f90e-44e4-b981-81e0f1237684", degreeCutoff:1, similarityCutoff:0.1, upperDegreeCutoff:2147483647, logProgress:true, nodeLabels:["*"],

Started streaming 1 record after 1 ms and completed after 3,187 ms.

4. Destacar os dez itens relacionados

g. Louvain

```
CALL gds.louvain.stream(
  'Customer_Similarity_Graph_Items',
  { relationshipTypes: ['Similar'], relationshipWeightProperty: 'score' }
)
YIELD nodeId, communityId
RETURN communityId, count(nodeId) AS size
ORDER BY size DESC;
```

Table RAW

communityId	size
4154	114
3321	31
4713	12
4229	6
4529	5
5062	5
6394	4
3642	4
4089	4
4601	4

Started streaming 4,583 records after 1 ms and completed after 1,193 ms.

h. Louvain Mutate

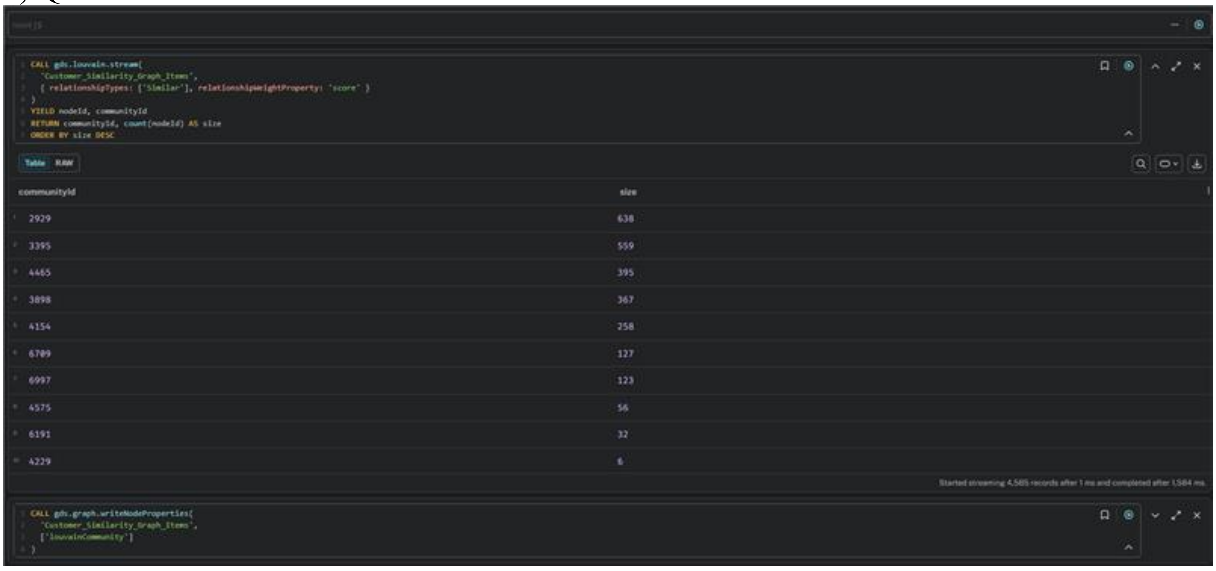
```
CALL gds.louvain.mutate(
  'Customer_Similarity_Graph_Items',
  {
    relationshipTypes: ['Similar'],
    relationshipWeightProperty: 'score',
    mutateProperty: 'louvainCommunity'
  }
);
```

Table RAW

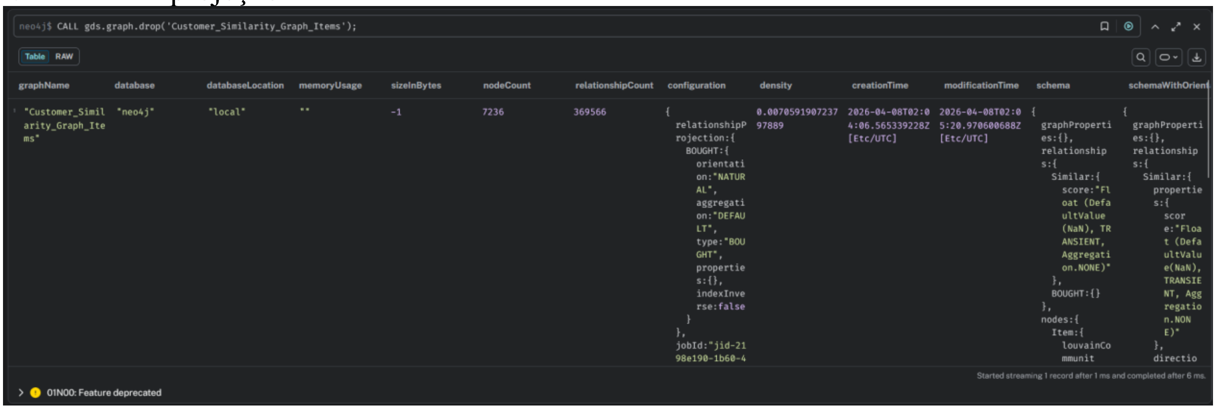
modularity	modularities	ranLevels	communityCount	communityDistribution	preProcessingMillis	computeMillis	postProcessingMillis	mutateMillis	nodePropertiesWritten	configuration
				9607073, p999:334 }						277-c999-453a-815d-7e6931deb9fe", forceSeedOptimization:false, logProgress:true, includeIntermediateCommunities:false, nodeLabels: ["*"], sudo:false, relationshipTypes:["Similar"], mutateProperty:"louvainCommunity", tolerance:0.0001 }

Started streaming 1 record after 1 ms and completed after 1,296 ms.

5) Quantas comunidades foram criadas com o total de itens em cada uma?



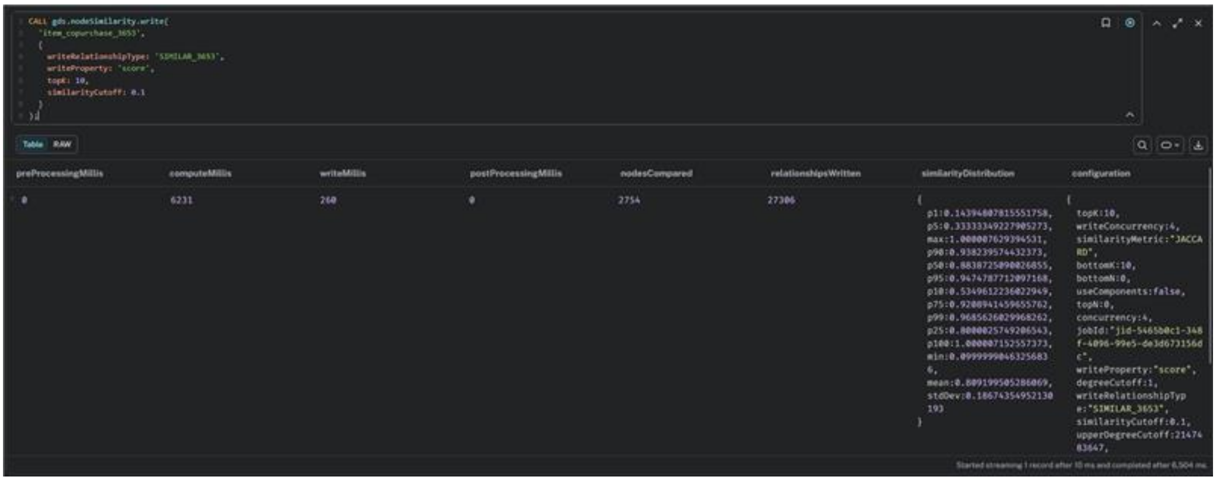
i. Eliminar a projeção



3) Item Similaridade

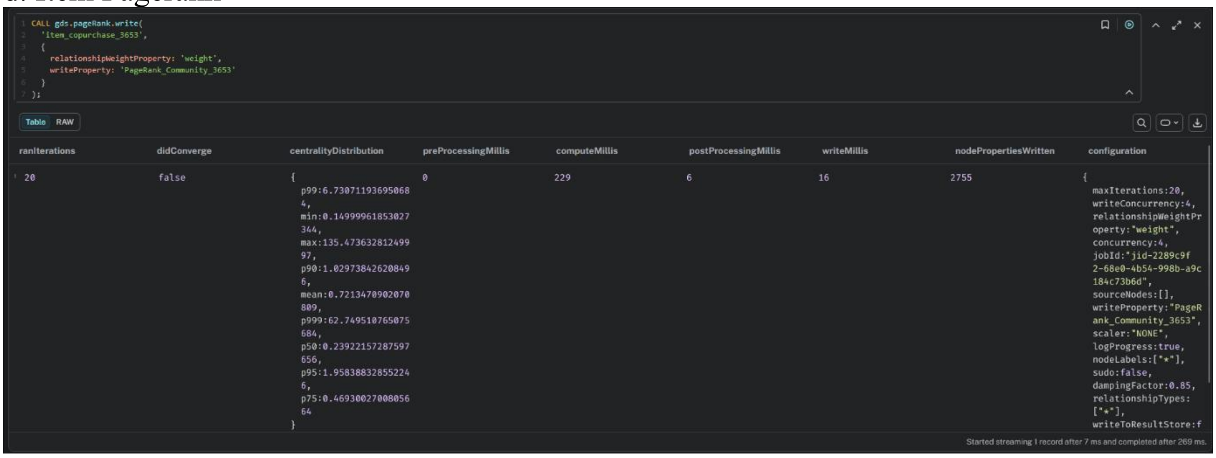
a. Item similaridade = 6115 (escolher comunidade existente em clientes)

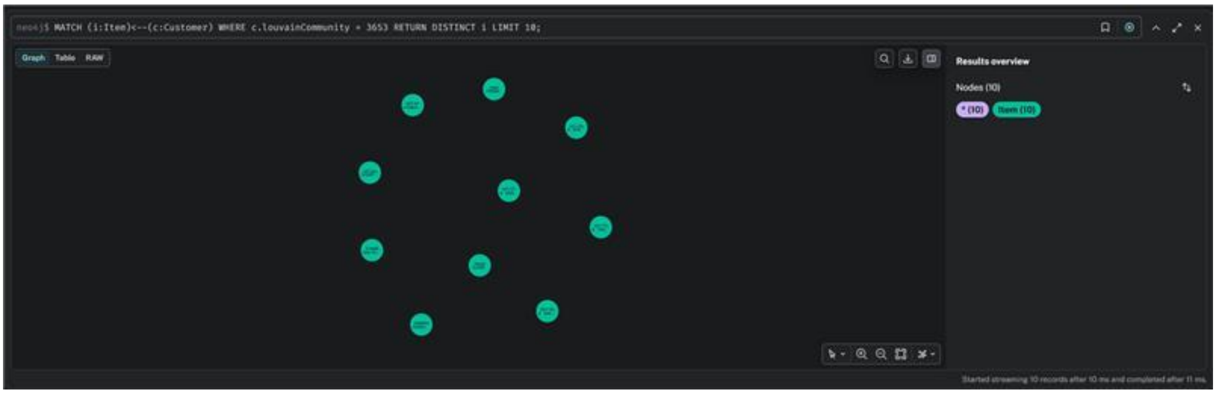
b. Gravar relacionamento similaridade



c. Item closeness

d. Item Pagerank






```
MATCH (i:Item)<--(c:Customer)
WHERE c.louvainCommunity = 3653
RETURN DISTINCT i.Description AS Item,
       i.PageRank_Community_3653 AS PageRank,
       i.Closeness_Centrality_Community_3653 AS Closeness
LIMIT 10;
```

Item	PageRank	Closeness
"FIRST CLASS LUGGAGE TAG "	0.1529782443580273	0.8169014084507042
"KEY FOB , BACK DOOR "	0.1609528142487669	0.9583333333333334
"KEY FOB , FRONT DOOR "	0.16002568874372385	0.96875
"KEY FOB , GARAGE DESIGN"	0.16186171280320674	0.9690721649484536
"KEY FOB , SHED"	0.16312067702399413	0.9693877551020408
"PINK CREAM FELT CRAFT TRINKET BOX "	0.16287568867948027	0.928
"SET OF 4 KNICK KNACK TINS DOILEY "	0.16714057002138263	0.9005524861878453
"VIP PASSPORT COVER "	0.16137106769113527	0.8766519823788547
"6 RIBBONS RUSTIC CHARM"	0.18072988930169046	0.9439252336448598
"EMBROIDERED RIBBON REEL EMILY "	0.162252104656317	0.8729508196721312

Started streaming 10 records after 49 ms and completed after 50 ms.

```
MATCH p=(i1:Item)-[r:SIMILAR_3653]->(i2)
WHERE r.score > 0.1 AND r.score < 0.8
RETURN i1.Description, i2.Description, r.score AS score
ORDER BY score DESC;
```

i1.Description	i2.Description	score
"MIRRORED WALL ART PHOTO FRAMES"	"HEART T-LIGHT HOLDER"	0.7998472116119175
"CERAMIC CAKE BOWL + HANGING CAKES"	"CAKE STAND WHITE TWO TIER LACE"	0.799837925445705
"COOKING SET RETROSPOT"	"COFFEE MUG PINK PAISLEY DESIGN"	0.7998266897746967
"COOKING SET RETROSPOT"	"COSY HOUR CIGAR BOX MATCHES "	0.7998266897746967
"COOKING SET RETROSPOT"	"CORDIAL GLASS JUG"	0.7998212689901698
"ANGEL DECORATION PAINTED ZINC "	"4 WILDFLOWER BOTANICAL CANDLES"	0.7998051631758403
"SET OF 4 DIAMOND NAPKIN RINGS"	"SET OF 2 TINS JARDIN DE PROVENCE"	0.7997724687144482
"LANDMARK FRAME CAMDEN TOWN "	"LANDMARK FRAME OXFORD STREET"	0.799752781211372
"WRAP MAGIC FOREST "	"WRAP DOLLY GIRL"	0.7997076023391813
"I LOVE LONDON WALL ART"	"LARGE CAKE STAND HANGING HEARTS"	0.7995965030262273

Fetch limit hit at 5,000 records. Started streaming after 51 ms and completed after 97 ms.

Encontrar similares

```
neo4j$ CALL gds.nodeSimilarity.stream('item_copurchase_3653', { topK: 1 }) YIELD node1, node2, similarity RETURN gds.util.asNode(node1).Description AS Produto1, gds.util.asNode(node2).Description A
```

Produto1	Produto2	similarity
"WHITE BEADED GARLAND STRING 20LIGHT"	"WRAP WEDDING DAY"	1.0
"WRAP WEDDING DAY"	"WHITE BEADED GARLAND STRING 20LIGHT"	1.0
"LARGE BLACK DIAMANTE HAIRSLIDE"	"MONTANA DIAMOND CLUSTER EARRINGS"	0.9988662131519275
"MONTANA DIAMOND CLUSTER EARRINGS"	"LARGE BLACK DIAMANTE HAIRSLIDE"	0.9988662131519275
"NUMBER TILE VINTAGE FONT 7"	"NUMBER TILE VINTAGE FONT No "	0.9977238239757208
"NUMBER TILE VINTAGE FONT No "	"NUMBER TILE VINTAGE FONT 7"	0.9977238239757208
"COPPER AND BRASS BAG CHARM"	"CRYSTAL SEA HORSE PHONE CHARM"	0.9949748743718593
"CRYSTAL SEA HORSE PHONE CHARM"	"COPPER AND BRASS BAG CHARM"	0.9949748743718593
"CROCHET BEAR RED/BLUE KEYRING"	"CROCHET DOG KEYRING"	0.994535519125683
"CROCHET DOG KEYRING"	"CROCHET BEAR RED/BLUE KEYRING"	0.994535519125683

Started streaming 10 records after 47 ms and completed after 6,809 ms.

6. Listar os itens com todos os scores calculados dentro da comunidade escolhida

```
MATCH (i:Item)<--(c:Customer)
WHERE c.louvainCommunity = 3653
RETURN DISTINCT i.Description,
       i.PageRank_Community_3653,
       i.Closeness_Centrality_Community_3653
ORDER BY i.PageRank_Community_3653 DESC
LIMIT 10
```

Table RAW

	i.Description	i.PageRank_Community_3653	i.Closeness_Centrality_Community_3653
1	"YELLOW COAT RACK PARIS FASHION"	135.47323222848922	0.8798722044728434
2	"WOOD BLACK BOARD ANT WHITE FINISH"	107.67536119900802	0.9290780141843972
3	"WOOD 2 DRAWER CABINET WHITE FINISH"	62.74926903949182	0.9395078605604922
4	"WICKER STAR "	38.493182132806105	0.932427843803056
5	"WICKER WREATH SMALL"	30.488475370657273	0.8836012861736334
6	"WICKER WREATH LARGE"	27.860888732865867	0.8930429128738622
7	"WHITE SPOT RED CERAMIC DRAWER KNOB"	26.93410076083781	0.9272911734866419
8	"WRAP WEDDING DAY"	26.103678508682975	0.772599663110612
9	"WHITE WICKER STAR "	26.010496297324217	0.9214237743451981
10	"WHITE SPOT BLUE CERAMIC DRAWER KNOB"	19.572855321689108	0.9222745625841184

Started streaming 10 records after 60 ms and completed after 72 ms.